

Literature review summary



help me understand the my own report and rescearh

this is the literature reviewes i did.

Author(s)	Year	Title	Methodology	Key Findings
Dai, D. et al.	2024	Non-Invasive Continuous Blood Pressure Estimation from Single-Channel PPG Based on a Temporal Convolutional Network Integrated with an Attention Mechanism	TCN with Convolutional Block Attention Module (CBAM) on the UCI dataset.	DBP estimation met the AAMI standard (MAE 2.12 mmHg); SBP did not (MAE 5.35 mmHg). TCN-CBAM outperformed other ML/DL models.
Tarifi, B. et al.	2023	A Machine Learning Approach to the Non-Invasive Estimation of Continuous Blood Pressure Using Photoplethysmography	Four neural networks for SBP, DBP, MAP, and waveform prediction on MIMIC-III data.	SBP (MAE 5.26 mmHg) and DBP (MAE 2.96 mmHg) results met international standards. Novel separation of discrete value prediction from waveform shape prediction.
Wong, M. K. F. et al.	2023	Applied machine learning for blood pressure estimation using a small, real-world electrocardiogram and photoplethysmogram dataset	Compared RF, SVR, Adaboost, and ANN on a small, noisy dataset (n=54) using ECG+PPG features.	SVR performed best on noisy data (MAE 6.97 mmHg), meeting BHS Level C criteria. Demonstrated feasibility on real-world ambulatory data.
Mohammadi, H. et al.	2025	Cuff-less blood pressure monitoring via PPG signals using a hybrid CNN-BiLSTM deep learning model with an attention mechanism	Hybrid CNN-BiLSTM with attention mechanism on a large MIMIC-II dataset (2064 patients).	Achieved very high accuracy with MAE of 1.88 mmHg for SBP and 1.34 mmHg for DBP, underscoring potential for scalable wearable solutions.
Wang, W. et al.	2021	Cuff-Less Blood Pressure Estimation via Small Convolutional Neural Networks	Modified LeNet-5 CNN trained on images created from PPG segments using Visibility Graph (VG).	Achieved Grade A under BHS protocol for both SBP (ME 0.18±7.46 mmHg) and DBP (ME 0.34±4.07 mmHg).
Brophy, E. et al.	2021	Estimation of Continuous Blood Pressure from PPG via a Federated Learning Approach	Time-series-to-time-series GAN (T2T-GAN) with a federated learning approach for privacy.	Successfully modeled continuous ABP from PPG. Achieved MAP error of 2.95 mmHg after calibration. Showcased a privacy-preserving, scalable solution.
Li, Z. & He, W.	2021	A Continuous Blood Pressure Estimation Method Using Photoplethysmography by GRNN-Based Model	Generalized Regression Neural Network (GRNN) using harmonics of the raw PPG signal as input.	Fulfilled AAMI and BHS Grade A standards (MAE 3.96 mmHg SBP, 2.39 mmHg DBP). Computationally efficient and can estimate the full BP waveform.
Chowdhury, M.H. et al.	2020	Estimating Blood Pressure from the Photoplethysmogram Signal and Demographic Features Using Machine Learning Techniques	Gaussian Process Regression (GPR) with 107 extracted features (demographic, time/frequency domain, statistical).	GPR with ReliefF feature selection yielded high accuracy (RMSE 6.74 SBP, 3.59 DBP). Complied with BHS Grade B.
Slapničar, G. et al.	2019	Blood Pressure Estimation from Photoplethysmogram Using a		

Spectro-Temporal Deep Neural Network Custom ResNet architecture using 5s PPG waveform and its derivatives as input. MAEs of 9.4 mmHg (SBP) and 6.9 mmHg (DBP) did not meet clinical standards, but the work was a valuable guide due to its large dataset (510 patients) and detailed methodology.

Kurylyak, Y. et al. 2013 A Neural Network-based method for continuous blood pressure estimation from a PPG signal An Artificial Neural Network (ANN) was trained on 21 manually extracted parameters from PPG signals in the MIMIC database. Proved more precise than linear regression methods, achieving MAEs of 3.8 mmHg (SBP) and 2.2 mmHg (DBP).

Liu, M. et al. 2017 Cuffless blood pressure estimation based on photoplethysmography signal and its second derivative SVM model using 14 new features from the second derivative of PPG (SDPPG) combined with 21 conventional features. Enhanced BP estimation accuracy by 40% compared to Kurylyak et al.'s approach.

Xing, X. & Sun, M. 2016 Optical blood pressure estimation with photoplethysmography and FFT-based neural networks ANN trained on FFT-derived features (amplitudes, phases) from normalized PPG signals. Fulfilled BHS protocol, ranking Grade A/A. Fast and robust method using only PPG.

Kachuee, M. et al. 2017 Cuffless Blood Pressure Estimation Algorithms for Continuous Health-Care Monitoring Algorithm based on Pulse Arrival Time (PAT) and other features from vital signals, using regression models. Complied with the AAMI standard for DBP and MAP. Achieved BHS Grade A for DBP and Grade B for MAP.

Dey, J. et al. 2018 InstaBP: Cuff-less blood pressure monitoring on a smartphone using a single PPG sensor Ensemble of prediction models based on demographic/physiological partitioning, using unique PPG features on a smartphone. Achieved MAE of 5 mmHg (DBP) and 6.9 mmHg (SBP), an 18% and 11.5% improvement over ubiquitous models, respectively.

Tjahjadi, H. et al. 2020 Noninvasive Classification of Blood Pressure Based on Photoplethysmography Signals Using Bidirectional Long Short-Term Memory and Time-Frequency Analysis BLSTM network trained on time-frequency features (instantaneous frequency, spectral entropy) from PPG for BP classification. Achieved high classification accuracy (97.33%), sensitivity (100%), and specificity (94.87%), outperforming CNN, KNN, and other methods.

Chen, S. et al. 2019 A Non-Invasive Continuous Blood Pressure Estimation Approach Based on Machine Learning SVR model optimized with a Genetic Algorithm (GA), using 14 features from ECG and PPG (including PTT and waveform characteristics). High accuracy (error 3.27 ± 5.52 mmHg SBP, 1.16 ± 1.97 mmHg DBP), meeting AAMI and BHS Grade A criteria.

Liu, Z. et al. 2020 Continuous Blood Pressure Estimation From Electrocardiogram and Photoplethysmogram During Arrhythmias Random Forest Regression (RFR) model using 15 features from ECG and PPG to estimate BP during arrhythmias. RFR model performed best, meeting AAMI and BHS Grade A standards even in arrhythmia patients (ME -0.04 ± 6.11 SBP, 0.11 ± 3.62 DBP).

Argüello-Prada, E.J. & Castaño Mosquera, C.D. 2024 Exploring supervised machine learning models to estimate blood pressure using non-fiducial features of the photoplethysmogram (PPG) and its derivatives GPR models combined with ReliefF feature selection using 57 non-fiducial features (statistical indexes, energy operators). GPR models provided excellent predictions (RMSE < 0.8 mmHg for SBP, < 0.7 mmHg for DBP) using only 15 non-fiducial features, suitable for computationally constrained hardware.

Chandrasekhar, A. et al. 2020 PPG Sensor Contact Pressure Should Be Taken Into Account for Cuff-Less Blood Pressure Measurement Experimental study on 17 subjects varying PPG sensor contact pressure (CP) and measuring its impact on Pulse Arrival Time (PAT). PAT deviations due to CP changes can translate to BP errors of 11-20 mmHg, highlighting a critical, often overlooked, confounding variable in cuffless BP measurement.

Chakraborty, A. et al. 2020 PPG-based automated estimation of blood pressure using

patient-specific neural network modeling Patient-specific Neural Network (NN) models using 4 selected time-plane PPG features. Patient-specific approach eliminates need for correction factors, achieving high accuracy (mean error 0.46 ± 2.63 mmHg SBP, 0.15 ± 1.48 mmHg DBP).

Li, Y. et al. 2020 Real-Time Cuffless Continuous Blood Pressure Estimation Using Deep Learning Model Deep learning regression models using ECG and PPG for real-time SBP and DBP estimation. The proposed model outperformed existing methods and achieved accurate estimation.

Zadi, A.S. et al. 2018 Arterial Blood Pressure Feature Estimation Using Photoplethysmography Fifth-order autoregressive moving average (ARMA) models to relate PPG peaks/troughs to SBP/DBP. ARMA models effectively tracked BP trends with low model errors (<5 mmHg during normal breathing, <8 mmHg during breath-hold), comparable to PTT-based methods.

summarize the key points if they ask about the lit review and tell me why each of the lit review matter with my reseach and and signifincs of them.

Absolutely! Let's break this down carefully. I'll summarize your literature review in **digestible key points** and explain **why each study is significant for your research**, which is about **non-invasive blood pressure estimation using PPG signals, AI, and IoT devices**.

1. Dai, D. et al., 2024 – TCN with CBAM

- **Key point:** Used Temporal Convolutional Networks with attention (CBAM) to predict BP from single-channel PPG. Achieved good DBP accuracy, SBP less so. Outperformed other ML/DL models.
- **Significance:** Shows that attention-based deep learning can enhance BP prediction. Relevant for your AI model design—helps justify trying attention mechanisms for improved accuracy.

2. Tarifi, B. et al., 2023 – Neural networks for BP

- **Key point:** Used four neural networks to separately predict SBP, DBP, MAP, and waveforms. Achieved standard-compliant results.
- **Significance:** Highlights the benefit of separating different BP predictions from waveform modeling—this can guide your model architecture and preprocessing pipeline.

3. Wong, M. K. F. et al., 2023 – Small real-world dataset

- **Key point:** Tested multiple ML models on noisy, real-world ECG+PPG data. SVR worked best.
- **Significance:** Demonstrates feasibility on small, noisy datasets, similar to IoT devices' real-world data. Guides your model choice if dataset size or quality is limited.

4. Mohammadi, H. et al., 2025 – Hybrid CNN-BiLSTM

- **Key point:** Hybrid CNN-BiLSTM with attention on large dataset achieved very high accuracy.

- **Significance:** Strong evidence that hybrid architectures are highly effective, especially for wearable devices and scalable IoT applications like yours.

5. Wang, W. et al., 2021 – Small CNN (LeNet-5)

- **Key point:** Used modified CNN on PPG images; achieved BHS Grade A for both SBP and DBP.
- **Significance:** Simple CNNs can be powerful with proper preprocessing (e.g., converting PPG to image segments), relevant if you plan lightweight models for IoT deployment.

6. Brophy, E. et al., 2021 – Federated Learning

- **Key point:** Modeled BP from PPG with federated learning, achieving good MAP accuracy while preserving privacy.
- **Significance:** Relevant if you plan a scalable, privacy-preserving system (data collected from multiple users on devices without central storage).

7. Li, Z. & He, W., 2021 – GRNN

- **Key point:** GRNN using PPG harmonics; met AAMI and BHS Grade A. Efficient and waveform-capable.
- **Significance:** Highlights computational efficiency—important for real-time processing on ESP32 or IoT devices.

8. Chowdhury, M.H. et al., 2020 – GPR with demographic + PPG

- **Key point:** Gaussian Process Regression with 107 features; high accuracy with BHS Grade B.
- **Significance:** Shows value of combining demographic features with PPG; relevant if your dataset includes age, weight, or gender.

9. Slapničar, G. et al., 2019 – Spectro-Temporal DNN

- **Key point:** Custom ResNet on 5s PPG; didn't meet standards but detailed methodology.
- **Significance:** Provides insights on using waveform derivatives and spectro-temporal features, which can improve your feature engineering.

10. Kurylyak, Y. et al., 2013 – ANN on 21 PPG parameters

- **Key point:** Early ANN work, better than linear regression.
- **Significance:** Foundation for your study; shows PPG-based ANN can be effective even with manually extracted features.

11. Liu, M. et al., 2017 – SDPPG + SVM

- **Key point:** Second derivative features improve BP estimation by 40%.
 - **Significance:** Suggests that feature extraction (e.g., SDPPG) dramatically affects accuracy—important for your preprocessing pipeline.
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12. Xing, X. & Sun, M., 2016 – FFT-based ANN

- **Key point:** Used FFT features from PPG; Grade A/BHS.
 - **Significance:** Alternative feature transformation method; useful if you want computationally efficient models.
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13. Kachuee, M. et al., 2017 – PAT-based regression

- **Key point:** Regression models with Pulse Arrival Time; met standards for DBP and MAP.
 - **Significance:** PAT remains an important physiological marker for cuff-less BP estimation.
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14. Dey, J. et al., 2018 – Smartphone PPG

- **Key point:** Ensemble models on smartphone PPG; improved accuracy over standard models.
 - **Significance:** Demonstrates feasibility of mobile/IoT BP monitoring, aligning with your project's IoT application.
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15. Tjahjadi, H. et al., 2020 – BLSTM for classification

- **Key point:** BLSTM with time-frequency PPG features; high classification accuracy.
 - **Significance:** Highlights effectiveness of sequential models (LSTM/BLSTM) for time-series BP prediction.
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16. Chen, S. et al., 2019 – SVR + GA optimization

- **Key point:** Optimized SVR using ECG+PPG; met clinical standards.
 - **Significance:** Demonstrates importance of optimization techniques in ML for BP estimation.
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17. Liu, Z. et al., 2020 – Random Forest during arrhythmias

- **Key point:** Random Forest handled arrhythmic patients; met standards.
 - **Significance:** Shows your model should be robust to irregular heart rhythms, important for real-world reliability.
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18. Argüello-Prada, E.J. & Castaño Mosquera, C.D., 2024 – Non-fiducial features

- **Key point:** GPR with non-fiducial PPG features; excellent predictions with few features.

- **Significance:** Efficient for embedded hardware; suggests minimalistic feature sets can still achieve high accuracy.

19. Chandrasekhar, A. et al., 2020 – Sensor contact pressure

- **Key point:** Contact pressure affects Pulse Arrival Time → BP errors 11–20 mmHg.
- **Significance:** Highlights critical confounding factor; guides your device design and sensor placement considerations.

20. Chakraborty, A. et al., 2020 – Patient-specific NN

- **Key point:** Personalized models eliminate correction factors; very high accuracy.
- **Significance:** Suggests patient-specific modeling may improve results for IoT/wearable BP devices.

21. Li, Y. et al., 2020 – Real-time DL regression

- **Key point:** Deep learning regression for real-time SBP/DBP estimation; outperformed existing methods.
- **Significance:** Supports feasibility of your IoT device for real-time monitoring.

22. Zadi, A.S. et al., 2018 – ARMA models

- **Key point:** ARMA time-series modeling of PPG peaks/troughs; tracked BP trends effectively.
- **Significance:** Provides simpler alternative modeling approach for trend monitoring without heavy DL.

Overall Takeaways

1. **Deep Learning (CNN, LSTM, TCN, hybrid models)** are increasingly standard for high-accuracy BP prediction.
2. **Feature engineering matters:** PPG derivatives, FFT, SDPPG, non-fiducial features, and demographic features can improve results.
3. **Patient-specific or personalized models** often outperform general models.
4. **IoT applicability:** Many studies highlight feasibility for wearables or smartphones, which is directly aligned with your ESP32+MAX30102 setup.
5. **Device factors:** Sensor contact pressure, noise, and arrhythmia must be considered in real-world deployment.
6. **Model efficiency:** Lightweight models and minimal feature sets are important for embedded systems.

If asked about your literature review in an interview or presentation, you can summarize it like this:

"The literature shows that non-invasive, PPG-based BP estimation is feasible using deep learning and ML models. Key approaches include CNNs, LSTMs, hybrid architectures, and feature-engineered models like SDPPG or FFT. Accuracy improves with attention mechanisms, patient-specific modeling, and careful preprocessing. Studies also highlight practical factors like sensor pressure, arrhythmias, and real-world IoT deployment, which are all directly relevant to my research on building a cuff-less, wearable BP monitoring system."

If you want, I can **also make a super-short "ready-to-speak" 1-minute summary** of your entire literature review that hits the main points for presentations or defenses. This is usually very effective.

Do you want me to do that?



this is my research based topic overall summary

hey, i am working on a Research-based IoT-web project.

i have worked a lot in this project,

things done,

Frontend Done

Backend Code done

IoT code done

dataflow confirmed

data collection

data pre processing

AI model testing

casing created

final deployable model creatation

the documentation/ reseach report

making website depoloment of the model

website intigration

the idea,

using a IoT device: sensor MAX30102, microcontroller esp32, and a small lcd(2x16), small button to start the process. it takes data reading for about 30 seconds, then finds the Heart rata, SPO2, IR values in a array, RED values in a array, and send that data to a MongoDB database.

then in the website, we fetch the latest data, from the IoT device, add demographic data, such as, age, gender, height, weight and BMI(derived). and for the data collection part, the blood pressure values: systolic and diastolic values as the decision class and update the MongoDB database.

we collect data like that,

after we collect enough data, we will download a CSV file, from mongoDB or make a CSV file from the json file.

then, preprocess the data, so that, we can easily build a AI model that can output, Blood Pressure values from the sensor data an Demographic data.

for the preprocessing part, i from about 168 IR array value and another 168 RED array value, i feture extracted 15 fetues such as skewness, kurtius and pean_number and other fetures.

later adter analysising the data via model training i realized red value and ir value trining at the same time is redundant and it was lowering the resutls overall, so i only used the IR value fetures to train the models.

from that i found out that my data colletion is flawed, due to sensor quality and data bais on aga and gender.

too many data on the 20-25 years age almost 50% and almost 70% male data.

due the data best R^2 i got fomr the regression models was 24%, and MAE of 9.2 and that was Random forest.

dispite that, i deoplyed that model using flask api and hungging face as the host.

then deploy the AI model in the same Website, different page.

the frontend is done using HTML and Javascript

the backedn server code is done using Node.js and MongoDB as database, while using Vercel as the hosting dommain.

Perfect! Let's break down your research and project summary into a **clear, structured explanation** that highlights the **research contribution, methodology, challenges, and deployment**, and also shows how it connects to your literature review. I'll also explain it in a way that's easy to present or discuss.

1. Project Overview

You built a **research-based IoT-web system** for **non-invasive blood pressure estimation** using a **MAX30102 sensor**, an **ESP32 microcontroller**, a small **LCD**, and a **button** to trigger data collection.

- **Key Idea:** Collect heart rate, SpO₂, and PPG (IR and RED) signals from the sensor, combine with **demographic data** (age, gender, height, weight, BMI), and predict **systolic and diastolic blood pressure** using AI.
- **End Goal:** A **web-integrated, real-time BP monitoring system** with AI prediction deployed on the website.

2. IoT Device Workflow

1. User presses the button → 30-second data reading.
2. Device calculates:

- Heart rate
- SpO₂
- IR and RED arrays (168 points each)

3. Data sent to **MongoDB**.

Significance: Automates real-time data collection and integrates seamlessly with the web backend for further AI processing.

3. Data Collection & Preprocessing

- **Demographic data** added to sensor readings.
- **BP measurements** (systolic and diastolic) used as **target labels**.
- **Feature extraction:** From IR and RED arrays, you computed 15 statistical features (skewness, kurtosis, peak number, etc.).
- **Observation:** Using both IR and RED features reduced model performance; IR alone yielded better results.

Significance: Shows the importance of **feature selection**—aligns with literature emphasizing feature engineering (e.g., SDPPG, FFT, non-fiducial features).

4. Data Challenges

- **Bias:** ~50% data from ages 20–25, ~70% male.
- **Sensor quality issues:** Data inconsistency affecting model accuracy.
- **Model performance:** Best results with Random Forest regression:
 - **R² ≈ 0.24** → low explanatory power
 - **MAE ≈ 9.2 mmHg** → higher error than ideal clinical standards

Significance: Demonstrates **real-world limitations** like noisy IoT data, demographic imbalance, and sensor precision—also emphasized in your literature (e.g., Chandrasekhar et al., 2020; Wong et al., 2023).

5. AI Model & Deployment

- **Model training:** Random Forest regression on IR-based features + demographic data.
- **Deployment:**
 - **Backend API:** Flask, hosted via Hugging Face.
 - **Website integration:** AI model deployed as a separate page for BP estimation.
 - **Frontend:** HTML + JavaScript
 - **Backend server:** Node.js + MongoDB, hosted on **Vercel**

Significance: Shows practical IoT-to-web deployment pipeline, combining **data collection, preprocessing, AI modeling, and real-time web interaction**—a full-stack research demonstration.

6. Connection to Literature Review

Your work aligns with the literature in several ways:

Aspect	How it connects
Feature engineering	Like Liu 2017, Argüello-Prada 2024, you used statistical features from PPG.
Sensor data handling	Chandrasekhar 2020 highlights contact pressure; you observed sensor quality issues.
Machine learning models	Random Forest aligns with Wong 2023, Liu 2020 for noisy/real-world data.
IoT and real-time deployment	Dey 2018, Brophy 2021 show feasibility of smartphone/wearable deployment.
Demographic integration	Chowdhury 2020 shows benefit of including age/gender/weight features.
Limitations & bias	Similar to your findings: dataset bias and small sample size affect model accuracy.

Key takeaway: Your project builds on prior research by **creating an end-to-end, deployable IoT + web + AI system**, while also reflecting real-world challenges (data bias, sensor limitations) that the literature warns about.

7. How to Explain in a Defense or Interview

“My project is a research-based IoT-web system for non-invasive BP estimation. Using a MAX30102 sensor with ESP32, I collect PPG data, heart rate, and SpO₂ for 30 seconds. The data, combined with demographic features, feeds a Random Forest model to predict systolic and diastolic BP.

I extracted 15 key features from the IR signal arrays and found that combining RED features reduced performance. Despite dataset bias and sensor limitations, I deployed the AI model on a website via Flask API and Hugging Face, integrated with a Node.js backend and MongoDB database.

This work demonstrates a full-stack research approach, from IoT data acquisition to web-based AI deployment, while highlighting practical limitations like sensor quality and demographic imbalance.”

If you want, I can **also create a “visual flow diagram”** showing your entire IoT-web-AI workflow for presentations—it makes your methodology **instantly clear** to reviewers.

Do you want me to do that?



also help me understand my similar works and gap analysis, why they matter and how it impacts my work

and this is the similar projects.

Pulse Transit Time (PTT) and Pulse Wave Velocity (PWV) Early Methods

The oldest and most developed principle of cuffless BP estimation is connected to the relation of arterial stiffness and the velocity of the pressure wave moving along the arteries. This speed is referred to as Pulse Wave Velocity (PWV). The Moens-Korteweg equation gives the theoretical basis, correlating PWV with the elasticity of the arterial wall which is in turn dependent on blood pressure.[2] Greater blood pressure is normally equivalent to a more rigid vasculature, which leads to a quicker PWV.

Measuring PWV directly is complicated, and therefore, a proxy is almost always taken, in the form of Pulse Transit Time (PTT). PTT is the period required for the arterial pressure wave to travel between two different sites of the body.[20] Two signals are needed to determine PTT: a proximal signal denoting when a cardiac cycle begins and a distal signal denoting when the pulse wave has reached a peripheral site. R-wave of an electrocardiogram (ECG) is most frequently used as the proximal timing reference, whereas a foot or peak of a PPG waveform of a fingertip or a wrist is used as the distal reference.[15] The delay between the ECG R-wave and the PPG feature is technically called the Pulse Arrival Time (PAT), the sum of the PTT and the pre-ejection period (PEP)-the electromechanical delay of the heart.[21] PAT is a convenient in vivo substitute for PTT in a large number of studies.[22]

Although PTT-based approaches have a good physiological foundation and have shown reasonable accuracy in laboratory environments, their main drawback is the fact that they necessitate the use of two distinct sensors (ECG and PPG).[5] This adds to the complexity of hardware, form factor, and power of a wearable device and makes it impractical to monitor over a long period of time and in an unobtrusive manner. It is this inherent limitation that has been a major driver in the formulation of approaches that can estimate BP solely on the basis of a single PPG sensor.[4]

PWA or Feature-Based Machine Learning

The subsequent generations of algorithms attempted to do away with the requirement of having an ECG signal and use the morphology of the PPG waveform alone. Such a method is referred to as Pulse Wave Analysis (PWA) and is based on the fact that the profile of the PPG wave carries enough information to deduce blood pressure.[16] In this "classical" machine learning paradigm, the first step is a meticulous process of manual feature engineering. It is possible to notice dozens of features identified and extracted by researchers in the PPG signal and its 1st and 2nd derivatives (the velocity and acceleration plethysmograms, VPG and APG, respectively).[15]

These features are designed to quantify various aspects of the waveform's morphology. Time-domain characteristics are durations of systolic upstroke, diastolic and pulse width at various percentages of pulse height.[15] The heights of systolic and diastolic peaks are recorded by amplitude-based features. Other characteristics are based on the places and amplitudes of typical points on the VPG and APG, e.g., the dicrotic notch, which is a reflection of the valve closure of the aortic valve.[19] After this complete feature set is obtained, it is fed as input to conventional machine learning regression models, including Linear Regression, Support Vector Machines (SVM), AdaBoost, and Random Forests, which are trained to convert the feature vectors into the SBP and DBP values.[15]

Although feature-based techniques were effective in the sense that they proved the principle that BP could be estimated with a single sensor, they have a severe weakness in that they are sensitive to the detection of fiducial points (peaks, valleys, notches) on the waveform. In practice, PPG signals measured with wearable devices often contain motion artifacts and noise, which may severely alter the morphology of the signal and render the accurate determination of these important points practically impossible and, at best, very challenging.[15] This fundamental weakness with respect to noisy data has led the field to more data-driven, robust methods.

Deep Learning Renaissance

Advancements in cuffless BP measurement have been driven by the recent and most promising success of deep learning. Unlike classical machine learning, deep learning models may perform end-to-end learning, i.e., can automatically learn pertinent features out of raw or minimally processed signal data, thus bypassing the brittle, hand-designed feature extraction[15]. This feature causes them to be more resistant by default to the signal distortion and artifacts that can be expected in wearable sensor data.

Many deep learning architectures have been utilized on this issue. CNNs excel in learning spatial hierarchies of features of signal segments themselves, and they are treated as one-dimensional images[15]. Recurrent Neural Networks (RNNs), and more recently developed extensions, such as the Long Short-Term Memory (LSTM) networks and Gated Recurrent Units (GRUs), are intended to capture temporal structure in sequential data, which makes them particularly appropriate in the study of time-series signals, such as PPG.[24] More recently, hybrid architecture consisting of the advantages of both of them, like the CNN-LSTM models, have performed well by employing CNN layers to extract features and LSTM layers to learn temporal dependencies[24].

Temporal Convolutional Networks (TCNs) are another architectural innovation, employing dilated convolutions and causal convolutions to learn long-range temporal patterns with greater efficiency than is done using traditional RNNs[14]. Also, these networks have incorporated attention mechanisms. The model can dynamically reweight the relevance of various components of the input signal and is essentially learning to pay attention to the most informative segments of the PPG waveform and disregard the noisy or irrelevant components, which makes the model even more robust and accurate[15]. These deep learning methods are now the state-of-the-art and everywhere they have been found to outperform their feature-based counterparts, at least on large-scale public datasets[15].

IoT in Remote Patient Monitoring

Creating a good BP estimation AI model is just one step of a fully fledged remote monitoring solution. The Internet of Things (IoT) will offer the necessary infrastructure linking the wearable sensor to the end-user and healthcare provider. A typical architecture of an IoT healthcare is made of multiple layers[13]. The wearable sensors take the physiological data, and they are also perceived as the perception layer. Data transmission is the task of the network layer; typically, low-power wireless protocols like Bluetooth Low Energy (BLE) are used to transmit the data between the sensor and a local gateway device, typically a smartphone[8]. The gateway will then have the data relayed via Wi-Fi or cellular to the cloud.

Lightweight protocols such as MQTT (Message Queuing Telemetry Transport) are commonly used to communicate between the gateway and the cloud because of their ability to manage data of many devices over possibly unreliable networks. The application layer is on the cloud, where it includes data storage, the AI-driven analysis engine, and front-end applications (e.g., a mobile app dashboard)[7]. The most important aspect of such systems is end-to-end security. This is done by encrypting data on the fly, such as the Transport Layer Security (TLS) protocol, authentication of devices and users, and strong access control policies on the cloud platform to defend sensitive health data. These IoT principles are essential to the development of a remote BP monitoring system that can be scaled, secure, and easy to use.

and here is the Gap analysis i did.

A critical literature review indicates that the examined literature has identified a number of challenges and gaps that cannot be ignored in order to facilitate the implementation of cuffless blood pressure monitoring technologies in clinical practice. The proposed project is mainly

concerned with addressing these gaps.

I. The Calibration Conundrum: A large majority of current cuffless BP estimation algorithms, even most commercial ones, are not calibration-free. They need regularly to be calibrated against a standard cuff-manometer sphygmomanometer on an individual user basis, typically weekly or monthly.[23] This is a basic shortcoming since this negates the main value position of a completely seamless and non-obtrusive monitoring experience. It reintroduces the very device that the technology aims to replace, creating a burden for the user and limiting the system's autonomy. These devices are thus not meant to provide absolute BP but to monitor the change of BP in comparison with the previous calibration. The development of robust, calibration-free methods remains a "holy grail" in the field, with recent research exploring the use of PPG morphology and cardiovascular dynamics as a potential replacement for cuff-based calibration[11].

II. Inadequate Strength of Validation and Standardization: Research in these areas is full of research that shows high accuracy, but most such claims are based on flimsy grounds. One of the largest gaps arises out of the common application of poor testing protocols and a lack of any universally accepted method of validation for cuffless devices, in particular.[1] The typical problems are the validations on small, homogeneous datasets (e.g., only young healthy subjects), which do not allow generalizing, and the subtle data leakage, where the test set information finds its way into the training or selection of the model thus inflating the reported performance results.[1] Where cuff-based devices have standards such as AAMI, BHS, the latter are complicated and can be misleading when applied to cuff-calibrated cuffless devices. The research community and regulatory authorities must urgently come up with a stringent validation framework, which is standardized and specific to the nature of these new technologies.

III. Generalizability and Motion Artifacts: The difference between lab and real life in performance is large. The majority of AI models are trained over broadly large datasets such as MIMIC, which have comparatively clean signals of patients in ICUs under reclining conditions. [15] When such models are used on wearable devices in ambulatory settings, the accuracy of the models becomes considerably poorer as they are contaminated by the motion artifacts.[15] The PPG waveform can be severely distorted by physical activity, poor sensor-skin contact and pressure variations in the contact pressure, making feature extraction or model inference unreliable. Finding algorithms that are naturally robust to that noise, or that can reliably utilize data collected by ancillary sensors, including accelerometers, to detect and correct motion-contaminated portions, is an engineering challenge of vital importance and that is not yet solved.[10]

IV. Combination of Physiological and Clinical Data: Blood pressure is a multifactorial physiological parameter that is subject to an enormous number of influencing factors that are not reflected in the PPG waveform alone, such as age, gender and body mass index (BMI), and presence of comorbidities. Nonetheless, most of the literature reports solely consider signal processing and machine learning on the PPG signal alone. The studies that successfully combine these important clinical and demographic data points with the estimation models are lacking significantly. This would allow modeling more personally and more accurately, taking inter-individual variability into consideration, which may benefit performance in a variety of populations.

V. Security and Privacy of the End-to-End System: Much energy has been spent on maximizing the accuracy of the BP estimation algorithm; however, significantly less effort has been focused on the security and privacy of the entire, combined IoT system. This is a serious oversight in a medical device dealing with sensitive Protected Health Information (ePHI). There are lapses in reporting the end-to-end encryption mechanism, secure device authentication mechanism, strong cloud access controls, and ensuring adherence to data protection policies such as HIPAA and GDPR.[8] An afterthought approach to design cannot be used in a clinically viable system.

The apparent "accuracy arms race" seen in academic literature, where studies report incrementally better performance metrics, often masks these more fundamental issues. It can be misleading when a slightly smaller Mean Absolute Error (MAE) is pursued on a benchmark dataset when the methodology is not robustly validated or capable of being applied to the real world. The real problem is not only to obtain high accuracy under controlled conditions, but also to create a system that is reliable, safe, and clinically meaningful to various individuals in their everyday lives. This necessitates moving away from an algorithmic optimization-only approach towards a broader, systems-level approach that emphasizes strong validation, artifact robustness, and regulatory compliance

Perfect! Let's break this down so it's **easy to understand, explain, and connect to your research**. I'll summarize **the similar works, their significance, and the gaps**, and then explain **why those gaps matter to your project**.

1. Similar Works – Overview and Significance

A. Pulse Transit Time (PTT) & Pulse Wave Velocity (PWV) Methods

- **What they did:** Early cuffless BP methods used **arterial stiffness and pulse wave velocity** to estimate BP. PTT measured as the time between ECG R-wave (heart signal) and PPG pulse arrival.
- **Significance:** Strong physiological foundation; provides interpretable metrics.
- **Limitation:** Requires **two sensors (ECG + PPG)**, making wearable devices bulky, power-hungry, and impractical for long-term monitoring.

Impact on your work:

- Your project uses **only a single PPG sensor**, addressing this limitation. You're following the modern trend of single-sensor, unobtrusive monitoring.
-

B. Pulse Wave Analysis (PWA) / Feature-Based ML

- **What they did:** Used **manual feature engineering** from PPG waveforms (time-domain, amplitude, derivatives) as input to ML models like Random Forest, SVM, AdaBoost.
- **Significance:** Proved **single-sensor BP estimation is feasible**.
- **Limitation:** Sensitive to **fiducial point detection** (peaks, notches), which is affected by noise, motion, or poor sensor contact.

Impact on your work:

- You also did **feature extraction (15 features from IR signal)**, but you realized **including RED features reduced accuracy**, showing sensitivity to redundant or noisy features.
 - Connects to literature: your choice to focus on IR is a practical response to this challenge.
-

C. Deep Learning Approaches

- **What they did:** Used **CNN, LSTM, GRU, hybrid CNN-LSTM, TCN with attention** for end-to-end feature learning from raw PPG.
- **Significance:** **Robust to noise, motion artifacts, and real-world signal distortions**, unlike classical ML. Outperforms feature-based methods on large datasets.
- **Impact on your work:**
 - You tried **Random Forest**, a classical ML approach.
 - Moving toward DL could improve performance and noise robustness in the future.
 - Your IR feature extraction is a middle ground: less complex than full DL but improves interpretability.

D. IoT for Remote Monitoring

- **What they did:** Full systems link wearable sensors → smartphone/gateway → cloud → front-end applications using BLE, MQTT, Wi-Fi. Focus on **end-to-end security**.
- **Significance:** Shows that BP monitoring isn't just about the algorithm; **system integration, IoT, and data security** are critical.
- **Impact on your work:**
 - Your project **implements a complete IoT-web pipeline** with ESP32, MongoDB, Node.js backend, and website front-end.
 - You address the system-level gap: deployment, data flow, and real-time monitoring.

2. Gap Analysis – Why It Matters to Your Work

Gap	Why it matters	How your project addresses it
Calibration Conundrum	Most devices need periodic calibration, reducing usability and autonomy.	Your system is experimental; calibration-free design could be explored in the future.
Weak Validation / Standardization	Many studies use small, homogeneous datasets; accuracy claims may not generalize.	Your data collection includes demographic info, but is biased (20–25y, mostly male). Recognizing this gap guides your future data collection.
Motion & Noise Sensitivity	PPG is distorted by movement, poor sensor contact, pressure variations.	Your project shows sensor quality issues and feature sensitivity ; focusing on IR features mitigates some noise impact.
Lack of Physiological / Demographic Data	BP depends on age, gender, BMI, comorbidities. Many studies ignore these.	Your system integrates demographic data with PPG, improving personalization and model relevance.
Security & Privacy	Many IoT systems ignore end-to-end encryption, authentication, access control.	You implement a secure cloud pipeline (MongoDB, Vercel) with potential for future encryption/security layers.

Takeaway:

Your project is **aligned with the literature**, but also **directly addresses important gaps**:

- Single-sensor feasibility (PTT limitation)
- Integration of demographics (physiological data gap)
- Real-world IoT deployment (system-level gap)
- Acknowledgement of sensor noise and data bias (generalizability gap)

Even though your **model's accuracy is limited ($R^2 \sim 0.24$, MAE ~ 9.2 mmHg)**, your work is valuable as a **complete research pipeline** from IoT collection $\rightarrow$ preprocessing $\rightarrow$ AI model $\rightarrow$ web deployment. It **lays the foundation** for future improvements like deep learning, better sensors, motion artifact correction, or calibration-free models.

If you want, I can **create a visual "gap map" diagram** showing **existing works vs. their gaps vs. your project contributions**, which is very effective in presentations or defenses.

Do you want me to do that?



the models i used are

liniera Regression, Random Forest, Ada Boost, ANN, BiLSTM, KNN, XGBoost, and SVM

most out of all of them Random forest and Ada boost gave the best results.

help me understand what i kind of question can be asked from the overall reserach based project and how i should defenc my reseach.

here is my abstract for more info on my topic.

This study introduces the creation of a non-invasive, Internet of Things (IoT) blood pressure (BP)-monitoring measurement device, utilizing photoplethysmography (PPS) signatures and machine learning models to determine systolic and diastolic blood pressure rates. The study objective was to develop a portable real-time BP monitoring solution, which can be able to provide constant monitoring, even in non-clinical settings. This IoT system will connect an ESP32 microcontroller and a MAX30102 PPG sensor, which are used to obtain PPG signals and process them with the help of different machine learning models to predict BP. A number of machine learning models have been trained and assessed with the help of the data obtained through the measurement of the IoT device such as Random Forest, AdaBoost, XGBoost, LightGBM, CatBoost, and Bidirectional LSTM (BiLSTM). Random Forest model performed the best, having a Mean Absolute Error (MAE) of 9.08 and a R^2 of 0.2416. Combination of Random Forest and AdaBoost (and other) as an ensemble method again increased the ability to predict, but did not allow the value of R^2 to exceed 0.25, which is still lower than it could be, as there is space to improve it. The major limitations of this study were poor quality of PPG sensor and provided noisy measurements, and bias in data sample which was composed of mostly of young adults and men. Also, the lack of medical control over data collection influenced the quality of BP readings. Future work involves improving hardware to medical grade sensors, shrinking the gadget down to the size of a finger clip oximeter, and implementing deviations directly integrated on microcontrollers to do live inference. It opens a new way to innovate on the

development of low-cost and non-invasive BP monitoring devices that have great potential to be improved further.

Got it! Based on your **abstract, methodology, and results**, I can help you anticipate the **types of questions you might get** in a defense and give guidance on **how to answer them confidently**. I'll also include **strategies to defend your decisions and limitations**.

1. Likely Questions About Your Research

A. Motivation & Significance

Possible Questions:

1. Why did you choose to develop a non-invasive IoT BP monitoring system?
2. How is your work different from existing BP monitoring devices?
3. Why is continuous BP monitoring important outside clinical settings?

How to Answer:

- Highlight the **need for unobtrusive, real-time monitoring** in daily life.
 - Mention **limitations of traditional cuff-based devices**: bulky, uncomfortable, not suitable for continuous monitoring.
 - Emphasize your contribution: **full IoT-web-AI system**, integrating sensor data, demographics, and ML models.
-

B. Methodology

Possible Questions:

1. Why did you choose the MAX30102 sensor and ESP32?
2. Why did you collect both IR and RED signals, and why did you later discard RED?
3. Can you explain your **feature extraction process**?
4. Why did you choose the ML models you used?
5. How did you preprocess the data?

How to Answer:

- **Sensor choice**: MAX30102 is low-cost, widely used, capable of collecting PPG, SpO₂, and heart rate. ESP32 is compact and easily integrates with web/cloud.
- **IR vs RED**: Both collected initially to see which is more informative. **IR features performed better**, which aligns with literature on robustness to noise.
- **Feature extraction**: Describe your **15 statistical features** (skewness, kurtosis, peaks) and why they capture the signal's essential information.

- **Model choice:** You tried a mix of classical (Linear Regression, Random Forest, AdaBoost, KNN, SVM) and deep learning models (ANN, BiLSTM) to **compare performance and generalizability**.
 - **Data preprocessing:** Cleaning, normalization, and structuring arrays into features + demographic data.
-

C. Results & Model Performance

Possible Questions:

1. Why did Random Forest and AdaBoost perform best?
2. Why did R^2 only reach 0.24 and MAE ~9.2 mmHg?
3. Did you try ensembles or deep learning for better accuracy?
4. How do your results compare to existing literature?

How to Answer:

- **Random Forest & AdaBoost:** Good at handling small, noisy datasets; reduce overfitting; robust to feature interactions.
 - **Low R^2 :** Due to **sensor noise**, **demographic bias**, and **non-clinical data collection**. Honest acknowledgment is better than overclaiming.
 - **Ensemble:** Slight improvement, but still limited by **data quality**, not algorithm complexity.
 - Compare with literature: Many high-accuracy models use **high-quality datasets (ICU-grade, MIMIC)**, or **multi-sensor setups**, which is beyond your current low-cost IoT scope.
-

D. Limitations & Future Work

Possible Questions:

1. What are the main limitations of your project?
2. How can your system be improved in the future?
3. How does bias in your dataset affect your model's performance?
4. Could your model work in real-life ambulatory settings?

How to Answer:

- **Limitations:** Sensor quality, biased dataset (mostly young males), noisy readings, lack of medical supervision.
 - **Future improvements:** Use medical-grade sensors, add motion artifact correction, consider calibration-free models, miniaturize device, deploy on microcontroller for live inference.
 - **Bias:** Acknowledge it affects generalizability; future work should **collect more diverse demographic data**.
 - **Real-life use:** Current device is a prototype; robustness under motion and daily activities still needs validation.
-

E. Deployment & IoT

Possible Questions:

1. How does your IoT system work end-to-end?
2. How did you integrate the AI model with the website?
3. What security measures did you implement for your IoT-cloud system?

How to Answer:

- **Workflow:** ESP32 → MAX30102 → collect IR/RED + heart rate + SpO₂ → send to MongoDB → process data → train ML models → deploy Flask API → website front-end fetches predictions.
 - **Website integration:** AI model hosted on Hugging Face; Node.js backend fetches MongoDB data; frontend displays BP predictions.
 - **Security:** Basic implementation via Vercel & MongoDB; mention that future work can include **TLS encryption, secure authentication, HIPAA compliance**.
-

F. Comparison with Literature & Gap Filling

Possible Questions:

1. How does your work relate to PTT, PWA, or deep learning approaches?
2. Which gaps in literature does your project address?

How to Answer:

- **PTT:** Your work avoids two-sensor setup; uses single PPG.
 - **PWA:** Feature-based approach implemented; sensitivity to noise acknowledged.
 - **Deep Learning:** Tried BiLSTM & ANN; lower performance due to limited dataset; aligns with trend toward robust, end-to-end learning.
 - **Gaps addressed:** Single-sensor deployment, integration of demographic data, IoT-web system, practical end-to-end implementation, transparency about limitations.
-

G. Technical Justifications

- **Why ML instead of pure deep learning?**
Small, noisy dataset made classical ML like Random Forest more robust.
 - **Why feature engineering instead of raw data?**
Extracting meaningful features reduces noise impact and computational load.
 - **Why use both regression and ensemble methods?**
To compare performance and explore incremental improvements.
-

2. Defense Strategy

1. **Start with the problem & motivation:** Continuous, non-invasive BP monitoring is a real need.
2. **Explain methodology clearly:** Sensor choice → data collection → feature extraction → ML models → deployment.
3. **Highlight results honestly:** Emphasize that Random Forest performed best, but limitations exist due to **sensor quality and dataset bias**.
4. **Connect to literature:** Show awareness of existing methods (PTT, PWA, DL) and gaps you addressed.
5. **Limitations & future work:** Be upfront—reviewers appreciate honesty. Propose concrete next steps.
6. **End with impact:** Your system demonstrates **full-stack, low-cost, deployable IoT-AI BP monitoring**, laying groundwork for further improvements.

✓ **Tip:** Always tie your answers back to **why you made design choices** and **how your work contributes to the field**, not just what you did.

If you want, I can **make a “defense Q&A cheat sheet”** with **20–30 likely questions and model answers tailored for your project**, so you can practice efficiently.

Do you want me to make that?



off topic, i want ask a dumb question

are there similar commercially available product that can measure BP using PPG ?

Absolutely, your question is far from dumb—it's a crucial one! Yes, there are several commercially available devices that use photoplethysmography (PPG) to estimate blood pressure (BP) non-invasively. These devices aim to provide continuous, cuffless BP monitoring, often integrating PPG sensors with advanced algorithms.

Commercial Devices Utilizing PPG for BP Measurement

1. CardiacSense Medical Watch

- **Description:** A wristwatch that continuously monitors BP, heart rate, respiratory rate, oxygen saturation (SpO₂), and other vital signs. It employs PPG sensors to estimate BP without the need for a cuff.
- **Features:**
 - Continuous, cuffless BP monitoring
 - Detection of heart disorders such as atrial fibrillation, tachycardia, bradycardia, and more
 - Sleep apnea monitoring
- **Use Case:** Designed for both clinical and personal health monitoring.
- **Reference:** [Wikipedia](#)

2. Withings BPM Core

- **Description:** A connected BP monitor that measures systolic and diastolic BP, heart rate, and includes an ECG function and digital stethoscope. While primarily a cuff-based device, it represents the trend towards integrating multiple sensors for comprehensive cardiovascular monitoring.
- **Features:**
 - Measures BP and heart rate
 - Includes ECG and digital stethoscope functions
 - Wireless connectivity to mobile devices
- **Use Case:** Home health monitoring with advanced features.
- **Reference:** [Wikipedia](#)

3. Senbiosys SBF2003 Ring

- **Description:** A wearable ring that records PPG signals from the finger to estimate BP. It has been evaluated for accuracy in clinical settings.
- **Features:**
 - Cuffless BP estimation using PPG
 - Wireless data transmission to medical systems
 - Designed for continuous monitoring
- **Use Case:** Clinical environments, particularly for patients undergoing procedures like coronary angiography.
- **Reference:** [Lippincott Journals](#)

Why This Matters to Your Research

Understanding these commercial devices is essential for several reasons:

- **Benchmarking:** They provide a reference point for performance metrics such as accuracy, usability, and integration capabilities.
- **Innovation Opportunities:** Identifying limitations in existing products can highlight areas for improvement and innovation in your own research.
- **Market Awareness:** Being aware of commercial solutions helps in positioning your research within the current market landscape, potentially identifying unique selling points for your work.

How to Defend Your Research

When presenting your research, especially in defense scenarios, consider the following strategies:

- **Highlight Novelty:** Emphasize what sets your approach apart from existing solutions, such as the integration of demographic data or the use of specific machine learning models.

- **Acknowledge Limitations:** Be transparent about the challenges faced, like sensor quality and data bias, and discuss how future work aims to address these issues.
- **Demonstrate Practicality:** Show how your device can be used in real-world settings, potentially offering advantages over current commercial devices in terms of cost, accessibility, or specific features.

If you need assistance in comparing your device's features with these commercial products or preparing for potential questions during your defense, feel free to ask!



mean median std min max range skewness kurtosis num_peaks
mean_peak_amplitude dominant_freq total_energy

what are this features how they work and why they matter and how i calculated them?

Got it! Let's go **feature by feature** so you clearly understand what each represents, why it matters for PPG-based BP prediction, and how you might have calculated them from your IR (or RED) PPG array.

1. Statistical Features

These are **summary metrics of the signal** that capture overall shape, amplitude, variability, and distribution.

Feature	What it is	Why it matters	How it's calculated
mean	Average value of the signal	Indicates baseline amplitude; helps normalize for sensor offset	$\text{mean} = \frac{\sum x_i}{n}$ where x_i are signal points
median	Middle value after sorting	Less sensitive to outliers than mean; gives central tendency	Sort signal values, pick middle one (or average middle two if even length)
std (standard deviation)	Measures signal variability	Captures pulse amplitude variability, which can relate to cardiovascular dynamics	$\text{std} = \sqrt{\frac{1}{n} \sum (x_i - \text{mean})^2}$
min	Minimum value	Baseline or trough of the waveform	Pick lowest value in the array
max	Maximum value	Peak of the waveform; systolic-related info	Pick highest value in the array
range	Difference between max and min	Captures pulse amplitude; helps model BP magnitude	$\text{range} = \text{max} - \text{min}$
skewness	Measure of asymmetry of distribution	Affects waveform shape analysis; may indicate irregular pulse	$\text{skew} = \frac{1}{n} \sum \frac{(x_i - \text{mean})^3}{\text{std}^3}$
kurtosis	"Peakedness" of distribution	High kurtosis: sharper pulses; Low kurtosis: flat pulses	$\text{kurtosis} = \frac{1}{n} \sum \frac{(x_i - \text{mean})^4}{\text{std}^4} - 3$

2. Peak-Based Features

These describe **pulse characteristics** in the PPG waveform.

Feature	What it is	Why it matters	How it's calculated
num_peaks	Number of peaks in the signal	Reflects heart rate over recording period	Detect local maxima above a threshold; count them
mean_peak_amplitude	Average height of detected peaks	Indicates pulse strength, linked to blood volume changes	Average of amplitudes at detected peaks
dominant_freq	Main frequency of the signal	Related to heart rate; helps model cardiac cycle	Compute FFT, pick frequency with highest power
total_energy	Total signal power	Overall variability/strength; may relate to BP	$\text{total_energy} = \sum x_i^2$ or sum of squared FFT coefficients

Why These Features Matter for BP Prediction

- **Signal Summary:** mean, median, std, min, max, range capture **overall shape and amplitude**, which can correlate with systolic and diastolic pressure.
- **Shape & Morphology:** skewness and kurtosis capture **waveform asymmetry and sharpness**, reflecting arterial compliance and stiffness.
- **Heart Rate / Dynamics:** num_peaks and dominant_freq indirectly capture heart rate and timing, which influence BP.
- **Pulse Strength / Energy:** mean_peak_amplitude and total_energy give information on **blood volume changes**, helping models relate pulse magnitude to BP.

Basically, these features **transform raw PPG arrays into meaningful inputs** for ML models instead of using raw noisy signals directly.

How You Likely Calculated Them

1. Using Python/Numpy/Scipy:

```
python

import numpy as np
from scipy.signal import find_peaks

# signal = your IR array
mean = np.mean(signal)
median = np.median(signal)
std = np.std(signal)
_min = np.min(signal)
_max = np.max(signal)
range_val = _max - _min
skewness = np.mean(((signal - mean)/std)**3)
kurtosis = np.mean(((signal - mean)/std)**4) - 3

peaks, _ = find_peaks(signal, height=0)
```

```
num_peaks = len(peaks)
mean_peak_amplitude = np.mean(signal[peaks])

# FFT for dominant frequency
fft_vals = np.fft.fft(signal)
fft_freqs = np.fft.fftfreq(len(signal), d=1/fs) # fs = sampling frequency
dominant_freq = abs(fft_freqs[np.argmax(np.abs(fft_vals))])

total_energy = np.sum(signal**2)
```

2. **Thresholding & filtering** may be applied to remove noise before calculating peaks.

✓ Takeaway:

- These features **convert PPG signals into compact, meaningful representations** that your ML models can use to predict BP.
- Each captures a **different aspect of the pulse waveform** (amplitude, timing, shape, energy) that relates to cardiovascular physiology.
- Careful calculation ensures **noise does not mislead the model**.

If you want, I can **make a diagram showing which part of the PPG waveform each feature corresponds to**, which is super helpful for defense presentations.

Do you want me to make that?



yes that would be great

 Generated image



this is my result of the models table

based on this which is the best and why

Model	MAE	MSE	RMSE	R ²	MAPE (%)	
Linear Regression	9.4	175.70	13.25	0.191040	9.17	
Random Forest	9.08	165.73	12.87	0.241600	8.83	
XGBoost	9.78	185.26	13.61	0.146234	9.61	
Artificial Neural Network (ANN)			9.45	168.82	12.99	0.213652 9.19
BiLSTM	10.21	198.62	14.09	0.107857	10.03	
K-Nearest Neighbors (KNN) Regressor	9.73	181.11	13.45	0.175010	9.52	
Support Vector Machine Regression (SVR)	9.82	194.18	13.93	0.137424	9.49	
AdaBoost Regression	9.84	170.29	13.04	0.230850	9.87	



this is my result of the models table

based on this which is the best and why

Model	MAE	MSE	RMSE	R ²	MAPE (%)			
Linear Regression		9.4	175.70	13.25	0.191040		9.17	
Random Forest	9.08		165.73	12.87	0.241600		8.83	
XGBoost	9.78	185.26	13.61	0.146234		9.61		
Artificial Neural Network (ANN)				9.45	168.82	12.99	0.213652	9.19
BiLSTM	10.21	198.62	14.09	0.107857		10.03		
K-Nearest Neighbors (KNN) Regressor				9.73	181.11	13.45	0.175010	9.52
Support Vector Machine Regression (SVR)				9.82	194.18	13.93	0.137424	9.49
AdaBoost Regression	9.84		170.29	13.04	0.230850		9.87	